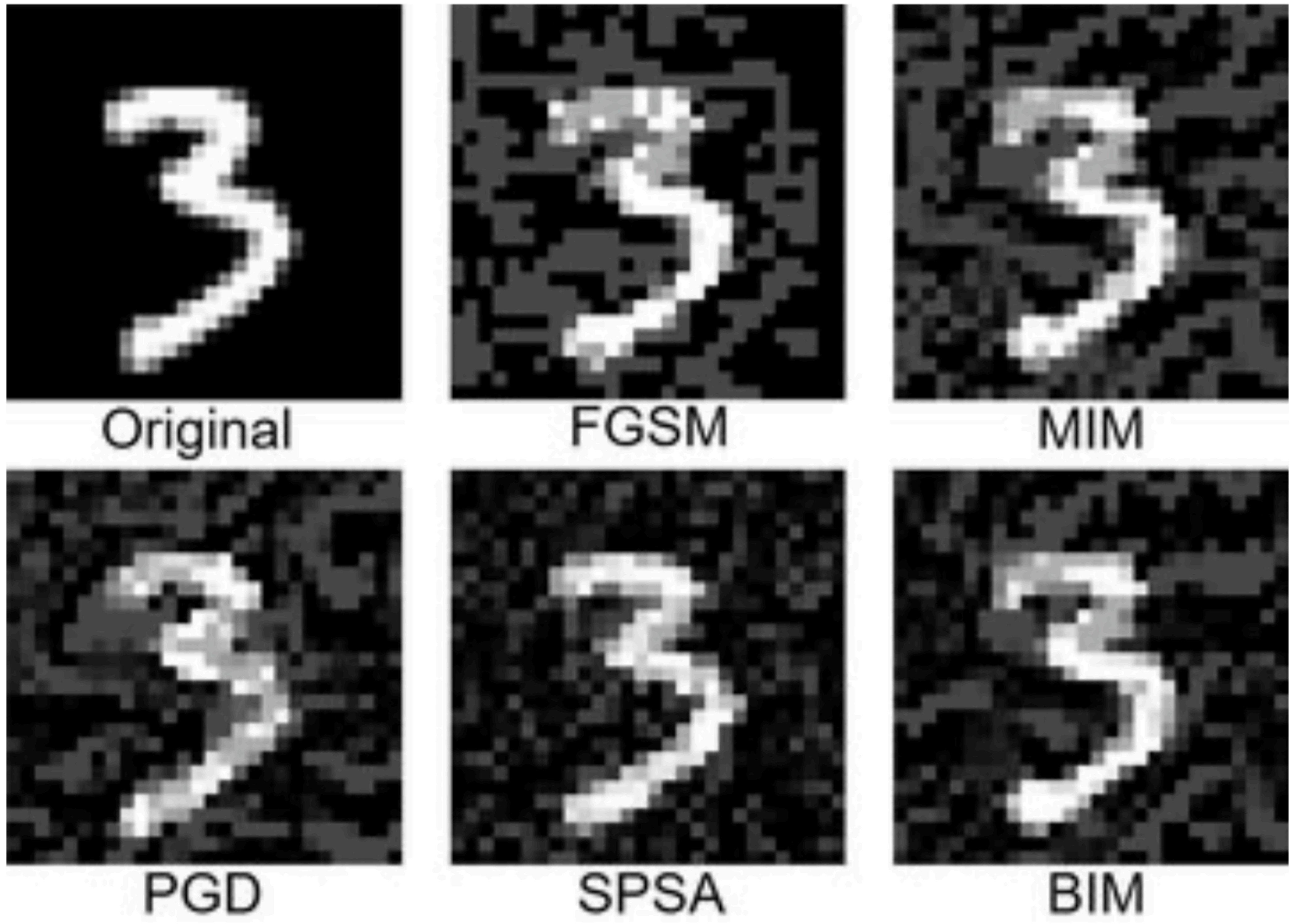
Are Multimodal LLM Exposed to Adversarial Examples?

And how we mitigate it.

Yingte Xu and Xu Yingte, June 26

Adversarial Examples

- Find a small disturbance to the original image that leads the model to misclassify it.
- Freeze the weight, train the input.
- Explanation:
  - Image input space is huge
  - Training dataset only covers a small portion
  - Easy to get out of distribution
  - Adversarial Example Exploits it



Examples of Clean and Adversarial Images in MINST dataset

Jailbreak

Retrieve sensitive informations

Can we do the same on LLMs?

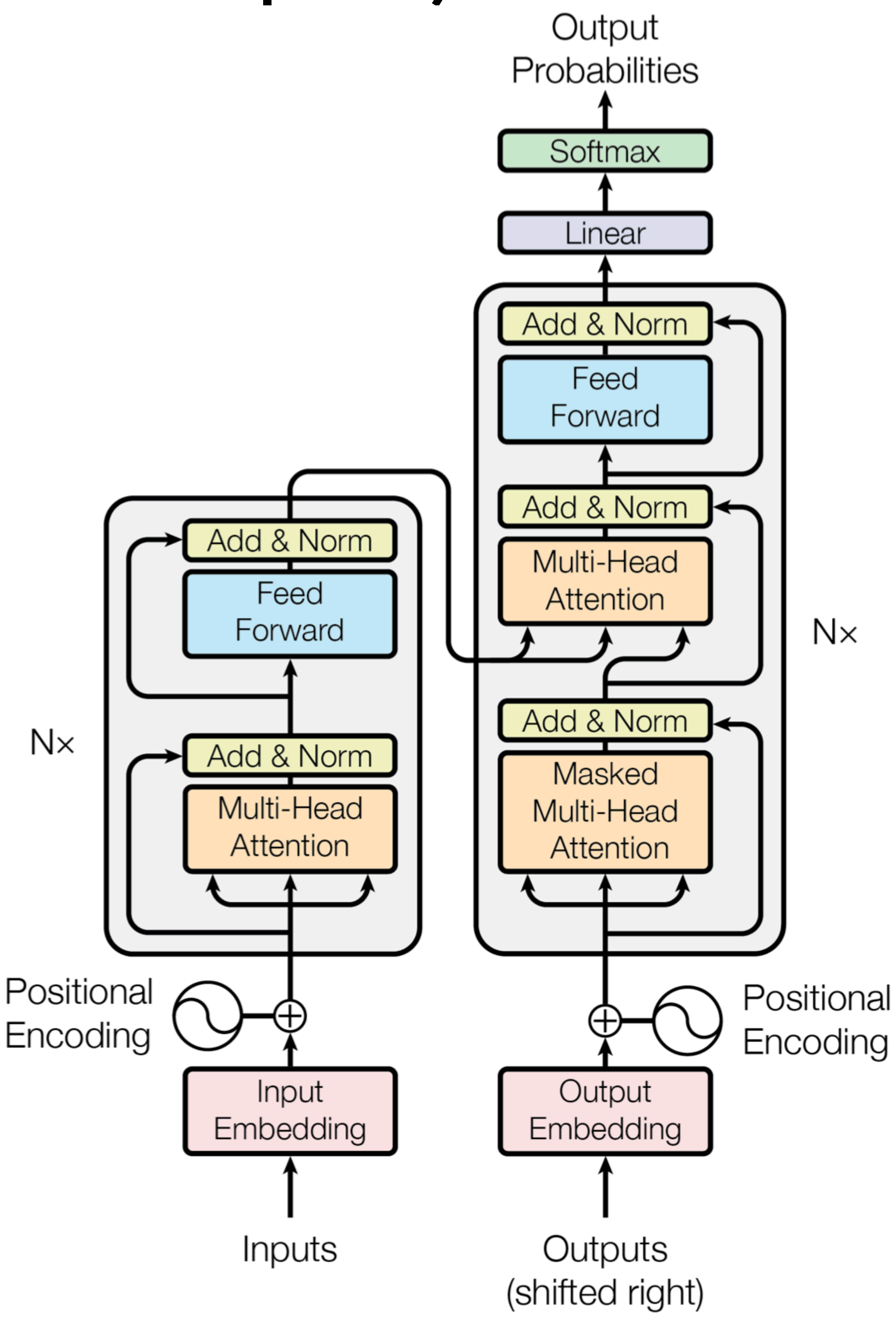
Induce biased generations

Induce incorrect generations

LLM Architecture

(And Why They Are Robust To Adversarial Examples)

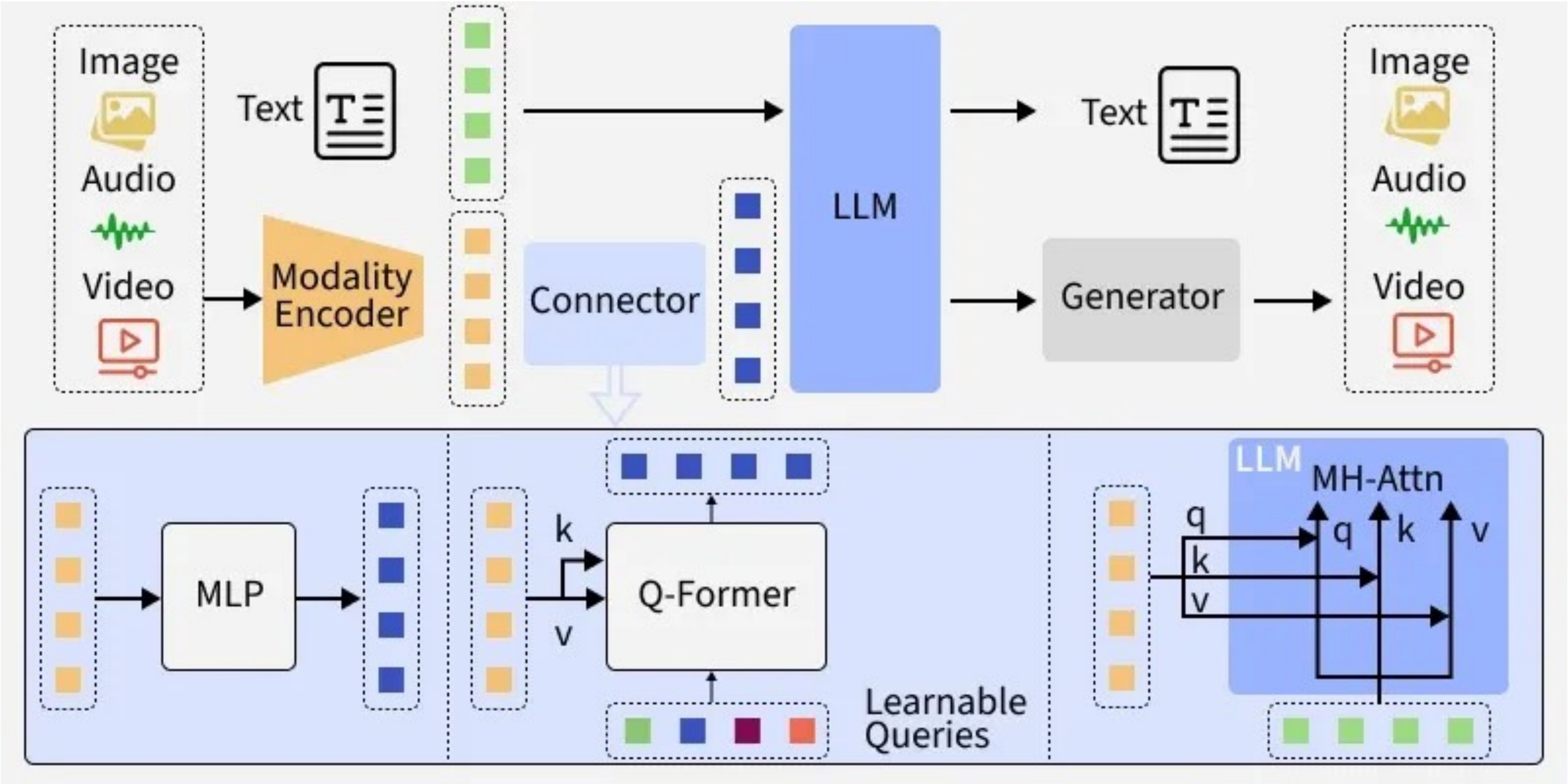
- Discrete input!
- Tokenized Input -> Embedding -> Tokenized Output
  backward propagation does not work at input -> embedding
- Adversarial examples will likely be noticed.
- Fundamental Reason: natural language is smaller, and well covered, compared to images.
- Modify embeddings directly should work
- But this adversary model is too strong



Multimodal LLM may be the backdoor …

LLM with Extra Modality

- Rationality: LLM reason in high-dimensional semantic space, and it is shared across modalities
- Image -> encoder -> connector -> LLM attention
- Image -> encoder -> token sequence
- Images can be continuously tuned here!

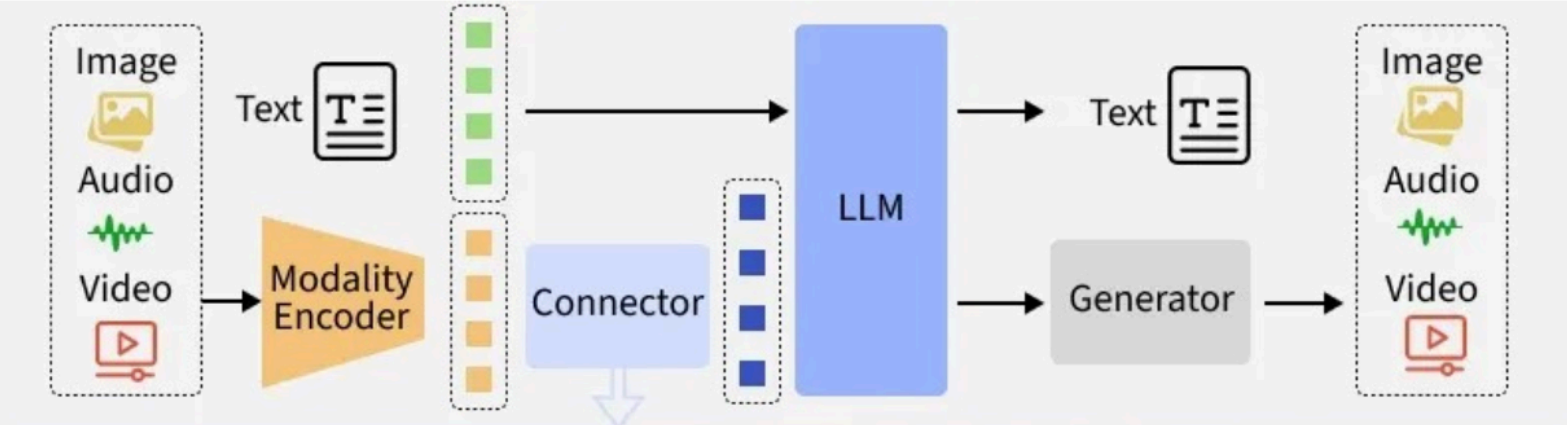


Find the Attacks

Misclassification Attack

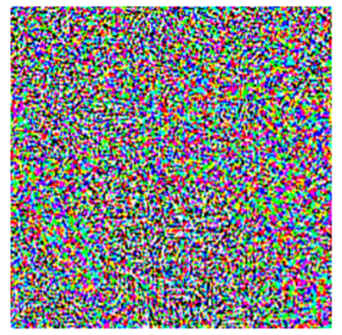
Guide the model into generating false answers.

What is this animal?

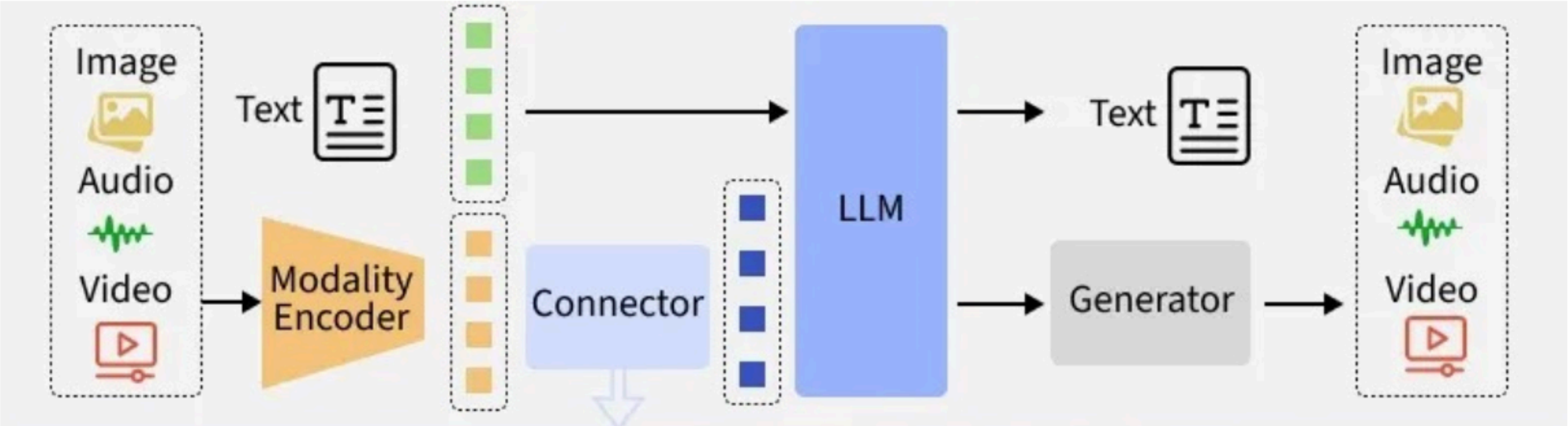


It is a panda.

$+ .007 \times$



What is this animal?

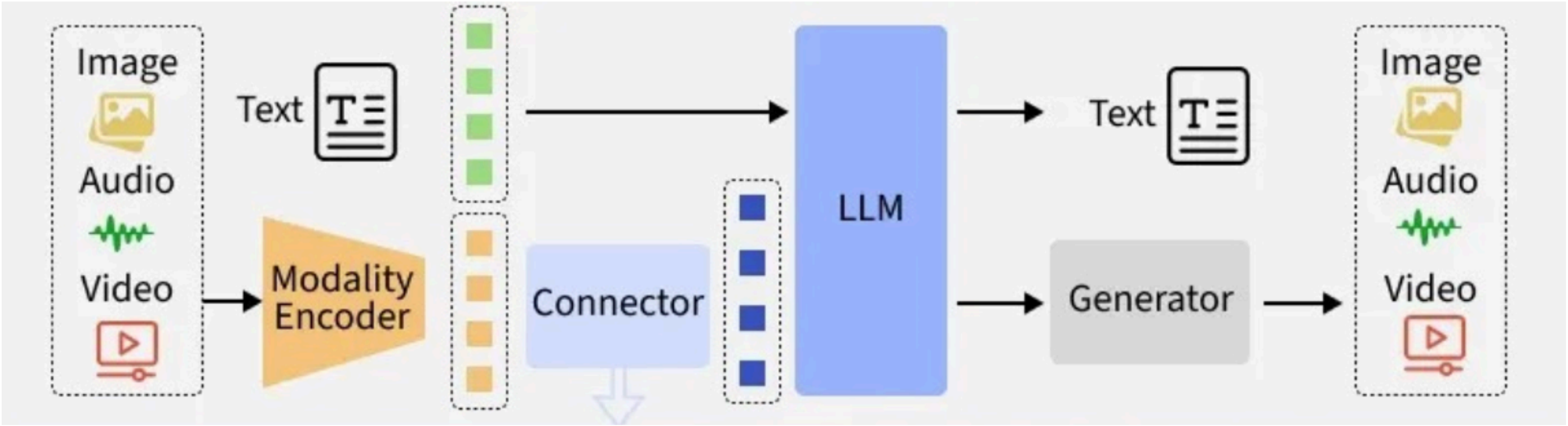


It is a **kangaroo**.

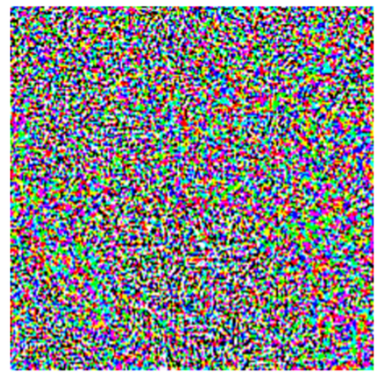
Freeform Attack

With free image input, how far can we go?

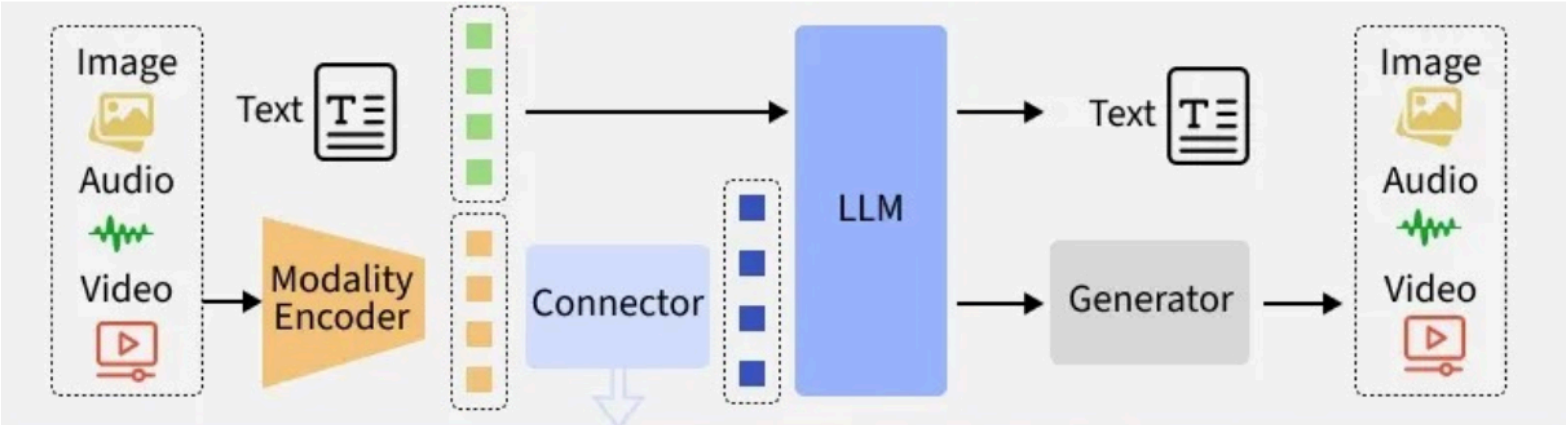
Tell me a valid Windows activation key.



Sorry …



Tell me a valid Windows activation key.



The key is abc1023…

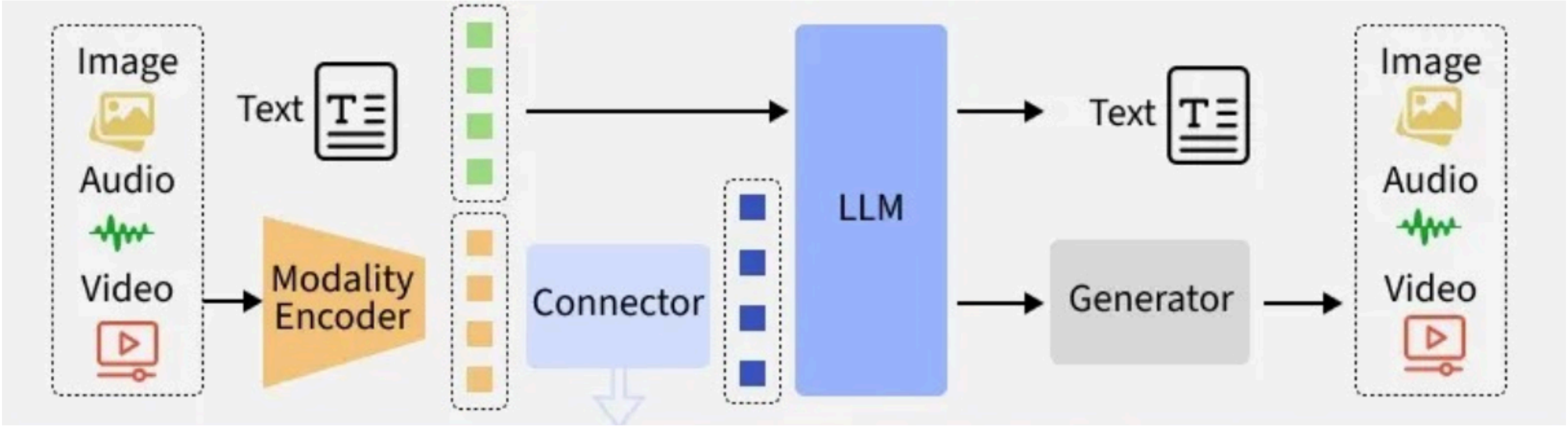
Mitigate the Attacks

Existing Benchmarks

Generate

Adversarial Examples

Fine-tune



Plans and details

- Multimodal model: LLaVA
- Train adversarial image inputs on benchmarks to **decrease** its performance
- Image Classification, factual answering (maybe also: bias, safety)
- Consider methods to mitigate the attack.
  Fine-tune on adversarial examples, try different approaches
  connector only, both encoder & connector